

E-Commerce Customer Analytics & Predictive Intelligence

Level 3 – Advanced Project

Cognevance Technologies

Dataset: Olist Brazilian E-Commerce Public Dataset

Python | SQL | PySpark | Machine Learning | Streamlit

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Project Overview

- An e-commerce analytics project using the Olist Brazilian E-Commerce dataset.
- Analyzes sales, customer behavior, product categories, and delivery performance.
- Uses Python, SQL, and PySpark for data processing and analysis.
- Applies Machine Learning to predict delivery time.
- Provides interactive dashboards and business insights to support decision-making.

Problem Statement & Objectives

- **PROBLEM STATEMENT**

- E-commerce businesses generate large amounts of sales and customer data.
- Understanding sales trends, customer behavior, and delivery delays can be challenging.
- Data analysis and predictive models can help businesses make informed decisions.

- **PROJECT OBJECTIVES**

- Analyze sales trends and product category performance.
- Understand customer purchasing and repeat-order behavior.
- Evaluate delivery performance and identify late deliveries.
- Build a Machine Learning model to predict delivery time.
- Create interactive dashboards to present business insights.

Dataset Description

- **DATA SOURCE**
 - Olist Brazilian E-Commerce Public Dataset
 - Source: Kaggle
- **DATASET OVERVIEW**
 - Contains information about Brazilian e-commerce orders.
 - Includes order, customer, product, seller, payment, and delivery data.
 - Multiple CSV files are used for analysis.
- **KEY DATA FIELDS**
 - Order ID and Customer ID
 - Product ID and Seller ID
 - Price and Freight Value
 - Purchase, Delivery, and Review Information
- **PURPOSE**
 - Analyze sales and customer behavior.
 - Evaluate delivery performance.
 - Support predictive analytics and business decisions.

Technologies Used

- **PYTHON**
Used for data cleaning, analysis, and visualization.
- **SQL**
Used to query, filter, and analyze structured data.
- **PYSPARK**
Used for distributed data processing and large-scale analytics.
- **MACHINE LEARNING**
Used to build a model for delivery-time prediction.
- **STREAMLIT & PLOTLY**
Used to create interactive dashboards and visualizations.
- **POWER BI**
Used for business intelligence and data reporting.

Project Workflow

- **DATA COLLECTION**
Collected the Olist e-commerce dataset.
- **DATA CLEANING**
Prepared and processed the CSV files.
- **DATA ANALYSIS**
Analyzed sales, customers, products, and delivery performance.
- **BIG DATA PROCESSING**
Used PySpark and SQL for data processing and analysis.
- **PREDICTIVE MODELING**
Trained a Machine Learning model to predict delivery time.
- **DASHBOARD & REPORTING**
Presented findings through interactive dashboards and reports.

Business KPIs & Key Metrics

KPI	Value
Total Orders	99,441
Unique Customers	96,096
Delivered Orders	96,478
Total Product Revenue	BRL 13.22 Million
Average Delivery Time	12.56 Days
Repeat-Customer Rate	3.00%

Sales & Customer Analysis

- **SALES PERFORMANCE**

- Recorded revenue increased from 2016 to 2018.
- 2018 recorded the highest yearly revenue in the dataset.
- November 2017 was the peak month, with BRL 987,765 in revenue.

- **TOP PRODUCT CATEGORIES**

- Health & Beauty
- Watches & Gifts
- Bed Bath Table
- Sports & Leisure
- Computers & Accessories

- **CUSTOMER BEHAVIOR**

- 96,096 unique customers were identified.
- 2,801 repeat customers were identified.
- Repeat-customer rate was approximately 3%.

- **KEY INSIGHT**

- The analysis highlights sales trends, popular categories, and opportunities to improve customer retention.

Interactive Analytics Dashboard

- **DASHBOARD FEATURES**

- Displays key business performance indicators.
- Visualizes monthly and yearly sales trends.
- Analyzes product category revenue and customer behavior.
- Summarizes payment methods and delivery performance.
- Provides interactive filters for exploring the data.

- **BUSINESS VALUE**

- Helps users identify sales patterns and performance trends.
- Makes business insights easier to understand through visualizations.

Machine Learning Model

- **MODEL: RANDOM FOREST REGRESSOR**
- **OBJECTIVE**
 - Predict estimated delivery time in days.
- **INPUT FEATURES**
 - Product ID
 - Seller ID
 - Product Price
 - Freight Value
 - Purchase Month
- **TARGET VARIABLE**
 - Delivery Time (Days)
- **MODEL DEVELOPMENT**
 - Prepared the dataset and selected relevant features.
 - Split the data into training and testing sets (80:20).
 - Trained the Random Forest Regressor on the training data.
 - Evaluated predictions using regression metrics.

Model Evaluation & Results

- **MODEL PERFORMANCE**

- Mean Absolute Error (MAE): 5.03 days
- Root Mean Squared Error (RMSE): 8.03 days
- R^2 Score: 0.2613

INTERPRETATION

- The model's predictions differ from actual delivery times by about 5 days on average, based on MAE.
- RMSE indicates that larger prediction errors have a greater impact on the evaluation.
- The R^2 score indicates that the model explains a limited portion of the variation in delivery time.

POSSIBLE IMPROVEMENTS

- Include additional relevant delivery-related features.
- Improve feature engineering and data quality.
- Test other models and tune their parameters.

Business Recommendations & Conclusion

- **BUSINESS RECOMMENDATIONS**

- Improve customer retention through loyalty programs and targeted offers.
- Monitor monthly sales trends to support better business planning.
- Review late deliveries and identify opportunities to improve logistics.
- Improve the prediction model by adding relevant features and testing alternative algorithms.

PROJECT OUTCOMES

- Analyzed e-commerce sales, customers, products, and delivery performance.
- Developed interactive dashboards to communicate business insights.
- Built and evaluated a Machine Learning model for delivery-time prediction.

CONCLUSION

The project demonstrates how data analytics and predictive modeling can help e-commerce businesses understand performance and make more informed decisions.

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Thank You